Department of Environmental Science and Engineering

Jatiya Kabi Kazi Nazrul Islam University

2nd Year 1st Semester: Final Examination, 2025; Session 2023-24

Course ID: ESE 2101; **Course Title:** Hydrology and Hydrogeology

Full Marks: 60; **Time: 03** (Three) Hours; **Credit: 03** (Three)

Answer any five questions out of eight

(Draw suitable sketches and diagram wherever necessary. The figures at the right margins indicate full marks)

1.	(a)	Define hydrological cycle. Explain with the help of a neat sketch the hydrological cycle in nature, indicating its various phases.	1+4
	(b)	What do you mean by Hydrology? Why is Hydrology important in engineering aspects?	1+3
	(c)	What is water balance? Derive the general hydrologic equation.	3
2	(a)	What is called Run off? How it is classified?	2+2
	(b)	Discuss the various factors which affect the Run off from a basin.	4
	(c)	Briefly explain the Run off Characteristics of streams.	4
3.	(a)	What is called precipitation?	1
	(b)	What are the mechanisms producing precipitation?	3
	(c)	Briefly explain the causes of precipitation?	4
	(d)	Briefly explain the various forms of precipitations.	4
4.	(a)	Define hydrograph and explain its components.	4
	(b)	Describe hydrograph separation and its importance.	3
	(c)	Explain the concept and applications of unit hydrograph in hydrology.	5
5.	(a)	Define Hydrogeology. What are the sources of groundwater?	1+2
	(b)	What is water profile? With the help of a neat sketch describe the groundwater profile.	1+4
	(c)	Outline top ten ways to preserve and protect groundwater. Explain the importance of groundwater in environment.	2+2
6.	(a)	Define Aquifer, Aquitard, Aquiclude, and Aquifuge.	4
	(b)	Explain the different types of aquifers with the suitable diagram.	5
	(c)	Illustrate the different factors which are related to a good aquifer.	3
7	(a)	Define evaporation and transpiration. Explain the factors affecting the evaporation process.	2
	(b)	Discuss briefly Analytical method of Evaporation.	6
	(c)	Illustrate the Dalton's law of evaporation.	4
8.		**Write short notes on the followings (Any three):**	3x4=12
	(a)	Darcy's law	
	(b)	Saltwater intrusion	
	(c)	Hydraulic conductivity and Hydraulic gradient	
	(d)	Specific yield and specific retention	

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Trishal, Mymensingh-2224, Bangladesh
2nd Year1st Semester B.Sc. (Eng.) Final Examination 2025; Session: 2023-2024
Course Code: ESE 2102 **Course Title:**Ecology-Lab
Total Mark: 60 Time Duration: 01.30 Hours; Date: 18/05/2026

Answer all the following questions. Sequence should be maintained in answering each question.

1. *Discuss the concept of food chains and food webs.* 6
2. Briefly explain how do you preserve ecological samples in your laboratory? 7
3. Explain energy flow in a pond ecosystem. 7
4. The dissolved oxygen (DO) content in a freshwater pond in JKKNIU campus was measured using the light and dark bottle method over a 24-hour period. The following data were recorded: Initial DO (D_0): 5.9 $mg \cdot L^{-1}$, Light Bottle DO after 24 h (D^L): 7.6 $mg \cdot L^{-1}$, Dark Bottle DO after 24 h (D^D): 4.8 $mg \cdot L^{-1}$, Calculate the Net Primary Productivity (NPP), Community Respiration (R), and Gross Primary Productivity (GPP) in $mg \cdot L^{-1} \cdot day^{-1}$. 10
5. During a field survey of the botanical garden in BAU Campus, the following number of individuals for each tree species was recorded: 10

Species	Number of Individuals
Mango	10
Jackfruit	6
Mahogany	7
Banyan	4
Neem	3

   - Calculate the total number of individuals (N).
   - Calculate D (Simpson's Diversity Index).

   Interpret whether the forest patch has low, moderate, or high diversity based on the result.
6. Viva Voce 20

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2nd year, 1st Semester; B.Sc. (Eng.) Final Examination, 2025; Session 2023-24
Course code: ESE-2103; Course Title: Oceanography and limnology

Total Marks: 60 **Time: 03:00 hours**

{Answer any 5 (five) questions out of eight. Draw suitable sketches and diagrams wherever necessary. The numbers at the end of the questions indicate full marks}

1.	a) Define oceanography? Narrate the different types of oceanography.	2+2
	b) What do you mean by the terms "thermocline, halocline, and pycnocline" with proper diagram?	4
	c) Briefly illustrate the ecological subdivisions of the sea.	4
2.	a) Explain the major dissolved salts in seawater and principle of constant proportions.	2+2
	b) Briefly discuss the geographical location of Bay of Bengal ecosystems with a suitable map.	4
	c) Describe the significant achievement of Bangladesh in the maritime areas against Myanmar and India.	4
3.	a) Mention the physicochemical properties of seawater. Identify the major ions of seawater.	2+2
	b) Write down the difference between continental and oceanic crusts.	3
	c) Discuss the major features of the ocean floor.	5
4.	a) Define the properties of a wave. Narrate the two types of wave phases.	2+2
	b) Explain the different types of waves with a diagram.	4
	c) Define tides. Discuss the major types of tides.	1+3
5.	a) What are coral reefs and their importance? Write down the different types of coral reefs.	2+3
	b) Explain the symbiotic relationships in coral reefs.	3
	c) Classify the ecosystem. Illustrate the ecosystem services.	1+3
6.	a) What do you mean by limnology?	2
	b) Describe the different types of inland water bodies.	5
	c) Define and classify the wetlands. Mention the common type of wetlands in Bangladesh, their local or Bengali names and identifying characteristics.	2+3
7.	a) Define lake stratification and describe seasonal pattern in temperate lakes.	1+3
	b) Explain thermal stratification and overturn in tropical lakes.	2+2
	c) Differentiate between dimictic, monomictic and polymictic lakes.	4
8.	a) Define NPP and GPP. Explain the methodologies for measuring primary productivity.	1+3
	b) Suppose a researcher takes a water sample from an ocean and performs the light and dark bottle method. The dissolved oxygen (DO) readings (in mg/L) are: Initial-DO: 8.0 mg/L; Light Bottle DO after incubation: 10.0 mg/L; Dark Bottle DO after incubation: 6.0 mg/L. Measure the respiration, NPP and GPP.	4
	c) Narrate the factors affecting primary productivity.	4

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2nd Year 1st Semester; B.Sc. (Eng.) Final Examination 2025; Session: 2023-24

Course Code: ESE 2107 Course Title: **Environmental Microbiology**

Total Marks: 60 Time: 03:00 Hours

Answer any **five** of the following questions. **Sequence** must be maintained in answering each question.

1.	a)	Give a Comparison between 'Microbiology' and 'Environmental Microbiology.'	3
	b)	Evaluate the importance of the study of Environmental Microbiology.	4
	c)	Briefly describe the scopes and applications of Environmental Microbiology.	5
2.	a)	Describe the microbial activities in soil with suitable example.	4
	b)	Explain the roleof microorganisms in nitrogen cycle.	4
	c)	Evaluate the importance of decomposers in soil.	4
3.	a)	Write down the major classes of microorganisms.	2
	b)	Describe the role of fungi in environmental science.	5
	c)	Compare the characteristics between bacteria and algae.	5
4.	a)	What is aquatic microbiology?	2
	b)	Explain the activities of microorganisms in a water with suitable example.	5
	c)	Describe the factors that influence microbial activity in water.	5
5.	a)	How does microbial activity influencewater quality? Explain with suitable examples.	6
	b)	Evaluate the relationship between microbial activity and soil nutrition.	6
6.	a)	Describe the functions of microorganisms in the atmosphere.	3
	b)	Explain the sources of bioaerosols with proper example.	5
	c)	Compare between bacterial and fungal aerosol?	4
7.	a)	Explain microbial interactions in environmental engineering with suitable examples.	4
	b)	Describe the positive interactions in microbial ecosystems.	4
	c)	Compare between mutualism and essential mutualism.	4
8.	a)	Define the term " Parasitism".	4
	b)	Explain the principles for control of microbial growth.	4
	c)	Describe the chemical methods of microbial control.	4

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2nd Year 1st Semester, B. Sc. (Eng.) Final Examination- 2025; Session: 2024-25

Course Code: ESE 2109 **Course Title:** Survey and Settlement
Total Marks: 60 **Time Duration :** 03:00 Hours

The figures in the margin indicate full marks. Answer any **Five** questions. Avoid split answering
Assume any reasonable value, if necessary. Symbols/Notations have their usual meanings

1 (a) What is surveying. Why it is important in engineering projects? 5

(b) Briefly discuss different types of surveying with their specific uses. 4

(c) Differentiate between Plane and Geodetic surveying. 3

2 (a) Explain different techniques of measuring the chain length when the stations are obstructed by applying principle of chain survey 6

(b) The area of a plot was measured by a defective engineering chain of 100.03 ft. The area was plotted down in a map to a scale of 1 inch = 100 ft and was calculated as 25.6 square inches. What is the true area of the plot? 6

3 (a) What is optical square? Prove that the line of object and eye sight remains in right angle for an optical square. 6

(b) Explain different methods of plane tabling.

4 (a) Write down the advantages and disadvantages of plane tabling.

(b) What is orientation? Describe the orientation procedure by back-sighting. 4

(c) Define curve. Explain different types of curves with figure. 5

5 (a) Discuss the errors that occurred during chain surveying. How do the errors can be eliminated? 4

(b) In a levelling work there are two setups of the instrument at A and B. From the position A staff readings on P, P_1, P_2, P_3, P_4 and C_p are 5.42, 7.21, 6.46, 5.38, 6.55 and 6.92 ft respectively. From position B staff readings on C_p, R_5, P_6, P_7, P_8 and P_9 are 8.78, 8.52, 6.24, 5.96, 6.35 and 7.54 ft respectively. If the R.L. of the bench mark P is 100′, calculate the reduced level of the above points by rise and fall method. 8

6 (a) Define leveling. What are the purposes of levelling? 3

(b) Discuss the effect of curvature on leveling. 4

(c) A line was measured with a Gunter's chain and found to be 3000 links. It was also measured with an Engineering chain and found to be 2100 ft. The Gunter's chain was 0.05 link too short. What was the true length of the engineering chain? 5

7 (a) Discuss the surveying works performed to complete a canal survey. 5

(b) The top of the Kutubdia light house is visible just above the horizon from a certain place in the Bay of Bengal. The distance of the light house from the observer is 20 miles. Calculate the height of the light house. If the height of the observer is 50 ft from the sea level. What is the height of the light house? 6

8 Write short notes (**any four**): 12

(i) Aerial Surveying (ii)Photographic surveying (iii)Closing error (iv) Methods of curve ranging (v) Dumpy Level

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2nd Year, 1st Semester B.Sc. (Eng.) Final Examination 2025; Session: 2023-24

Course code: **ESE 2111**; Course title: **Soil Mechanics**

Total Marks: **60**; Time duration: **03:00 Hours**

Answer any **five** of the following questions. **Sequence** must be maintained in answering each question. The figures at the right margin indicate full marks of the respective questions.

1. (a) Define soil mechanics and explain the relationship between environmental science and soil mechanics. **1+3**

(b) Briefly narrate the problems of soil associated with Civil Engineering. **4**

(c) Explain the role of soil in biosphere and hydrosphere. **4**

2. (a) What is pedosphere? Describe the soil composition **1+2**

(b) Illustrate soil horizon by explaining the composition and characteristics of each horizon. **5**

(c) Outline the soil formation factors in detail. **4**

3. (a) Why geotechnical engineers often use Phase Diagram? Illustrate Two and Three phase system. **3**

(b) Give a brief explanation of Unified Soil Classification System (USCS). **4**

(c) Soil has void ratio 0.72, moisture content 12%, specific gravity 2.72. Determine its dry and moist unit weight, and how much water need to be added to make it saturated. [γ_w=9.81 kN/m^3] **5**

4. (a) Discuss how soil texture, structure, and compactness affect soil porosity. **3**

(b) Discuss the detail procedure of determining Plasticity Index (PI) of soil **5**

(c) An undisturbed saturated clay specimen has volume of 18.7 cm^3 and a mass of 30.1 g. After drying in the oven, the mass reduces to 18 g. The volume of dry specimen is 9.7 cm^3. Now determine the water content in soil and the shrinkage limit. **4**

5. (a) What is stoke's law? Derive the stoke's law. **4**

(b) Give a brief explanation of Unified Soil Classification System (USCS). **4**

(c) For fine grain analysis, which method is used? Explain the technique. **4**

6. (a) Illustrate Mohr-Coulomb failure criterion. From which two sources do soil derive its shear strength? **4**

(b) Write advantage and disadvantages of CD, CU and UU Triaxial test. **5**

(c) Describe Unconfined Compression Test. **3**

7. (a) Explain the composition of soil air and atmospheric air. Why does soil air contain higher CO_2 and lower O_2 compared to atmospheric air? **4**

(b) Discuss the factors that influence the composition of soil air. **4**

(c) Describe the various factors that affect soil temperature. **4**

8. Short notes (any 3) **3x4**

(a) Atterburg limit

(b) Soil formation process

(c) SPT-N test

(d) Darcy's Law

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2nd Year 1st Semester B.Sc. (Eng.) Final Examination 2025; Session: 2023-2024

Course Code: **ESE 2113;** Course Title: **Statistics for Environment**

Total Mark: 60; Time Duration: 03.00 Hours

Answer any **five** of the following questions. **Sequence** must be maintained in answering each question.

1. a) What are the sources of statistical data? Write down the properties of good measures of central tendency. **2+2**

b) "Different types of means are used for different conditions"-Justify the statement with relevant examples. **3**

c) For the following dataset of ages of 20 employees: 23, 25, 27, 28, 30, 32, 35, 37,40, 42, 43, 45, 48, 50, 52, 55, 58, 60, 62, 65 construct a frequency table with class intervals of 10 years starting from 20-30, and calculate the mean, median and mode. **5**

2. a) What is measure of dispersion and what is the significance of measure of dispersion? **1+2**

b) Prove that, for a set of two unequal observations, each of mean deviation and standard deviation is half of the range. That is, MD=SD=R/2. **4**

c) The following data represent the marks of 5 students: 2,4,6,8, and 10. **2+3**

i. Calculate 2nd and 3rd central moment. $\rightarrow \mu_2 = \frac{\Sigma(x_i-\bar{x})^2}{n}$ & $\mu_3 = \frac{\Sigma(x_i-\bar{x})^3}{n}$

ii. Estimate the skewness and interpret the result. $= \beta = \frac{\mu_3}{\sqrt{\mu_2^3}}$

3. a) What is correlation? Distinguish between positive and negative correlation **3**

b) Prove that the value of correlation coefficient lies between -1 to +1 i.e. $-1 \leq r \leq 1$. **4**

c) The data gives the hours of sunlight and the growth (cm) of specific plant: **5**

Sunlight hours	5	6	6.5	7	7.5	8	8.5	9
Plant growth (cm)	8	8.5	8.3	11	11.5	14	15	15

i) Identify the independent and dependent variable in the data.

ii) Fit a linear regression of plant growth on sunlight hours and interpret the coefficients.

iii) Estimate the plant growth when sunlight hour is 8.8.

4. a) Define mutually exclusive event. State and prove the Total probability theorem. **1+4**

b) Mr. Jalal who works in Motijheel has two routes to go home: through the Ramna Park or Curzon Hall. He uses the Curzon Hall route one third of the time and the other route two thirds of the time. If he goes by the Ramna Park, he gets home by 6 o'clock 75% of the time; if he goes by the Cum Hall, he reaches home by 6 o' clock only 70% of the time, but he likes Curzon Hall campus better that way. One day he reached home after 6 o'clock. What is the probability that he used Curzon Hall route to go home? **4**

c) In a certain school, 20% of the students failed in English, 15% of the students failed in Mathematics and 10% of the student failed both in English and Mathematics. A student is selected at random. If he failed in Mathematics. What is the probability that he also failed in English? **3**

$P(CH) = \frac{1}{3}$

$P(RP) = \frac{2}{3}$

$P($

$P(A) = P\sum_{i=1}^{n} P(B_i)\,P(A|B_i)$

Department of Environment Science and Engineering

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2nd Year 1st Semester, Session: 2023-24; B. Sc. (Eng.) Final Examination- 2025

Course :**ESE 2108Engineering Drawing**, Total Marks: 60, Group: **B**, Time : 03:00 Hrs

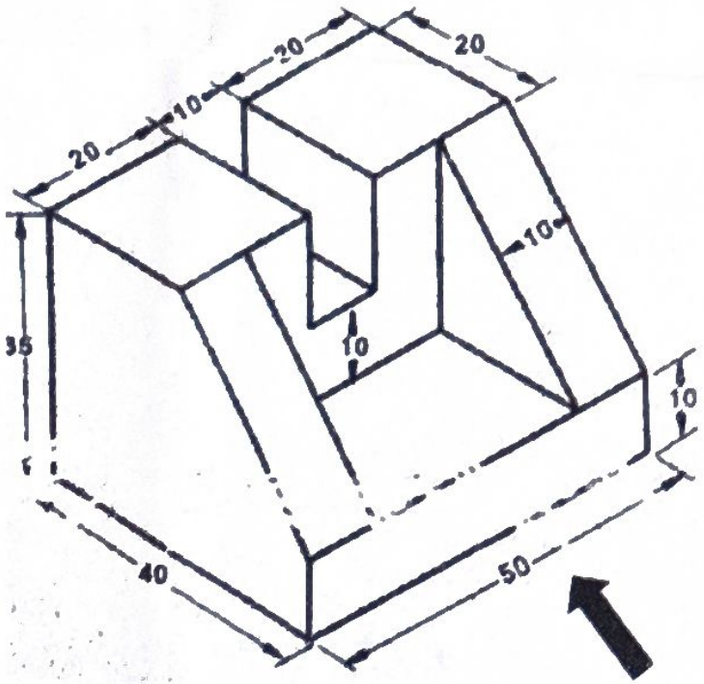
1. (a) What are the importances of engineering drawing? Write down the different drawing instruments with their specific function. 5

   (b) Draw a regular heptagon of 1 inch side. 5

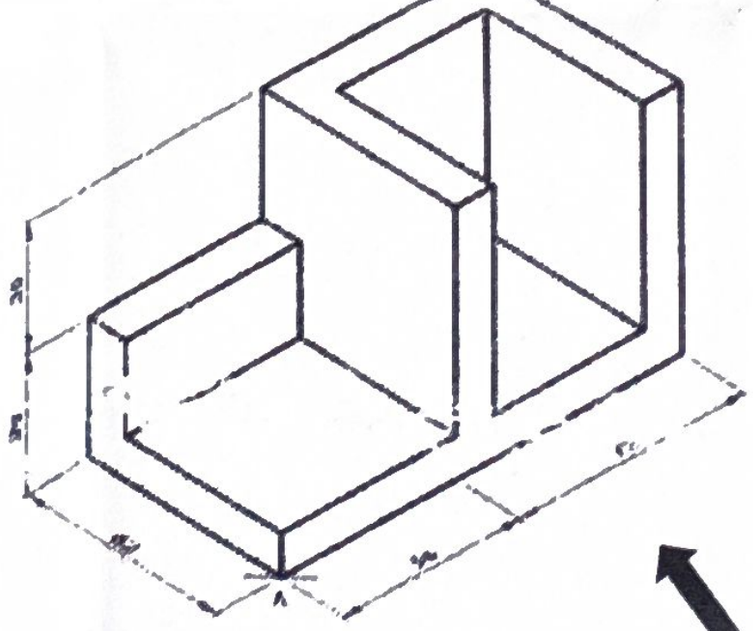
   (c) Draw an ellipse inside a rectangle of 4*2 inch size. 5

2. (a) Draw isometric view of a circular prism of 2 inch dimeter and 4 inch long lying on a plane parallel to horizontal surface. 10

   (b) Draw the orthographic views of Fig. (a) Dimensions are in mm and Fig. (b) Front from A 15



(a)



(b)

3. Viva-voce 20

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Trishal, Mymensingh-2224, Bangladesh

2nd Year 1st Semester; B.Sc. (Eng.) Final Examination 2025; Session: 2023-2024

Course Code: ESE 2105 **Course Title:** Ecology

Total Mark: 60 Time Duration: 03.00 Hours

Answer five of the following questions. **Sequence** must be maintained in answering each question. The figures at the right mark indicate full marks for the respective questions. (Draw suitable sketches and diagram wherever necessary.)

1. a) Define the terms: i) Ecology, ii) Autecology and iii) Synecology 4
   b) Differentiate between population and Community with example. 4
   c) Describe the application of Ecology in Environmental Science and Engineering. 4

2. a) What do you mean by ecosystem? Explain the functions of ecosystem. 1+2
   b) Discuss a pond ecosystem with proper diagram. 4
   c) Exemplify the role of energy flow in ecosystem. 5

3. a) What is Biogeochemical cycle? Discuss about the carbon cycle with flow diagram. 1+4
   b) Why Biogeochemical cycle is important for maintaining balanced ecosystem. 4
   c) Explain ecological pyramid with example. 3

4. a) Define species richness. 2
   b) Describe the Simpson's Diversity Index and its significance in biodiversity measurement. 5
   c) What are biodiversityindices,and why are they important in ecological studies? 2+3

5. a) Explain marine ecology with proper example. 3
   b) How is marine ecosystem affected by human activities? 4
   c) Illustrate the Productivity of an ecosystem with their types and equations. 5

6. a) What do you mean by Food web? 3
   b) Exemplify the classifications of food chain. 4
   c) Explain the food chain of a forest ecosystem with proper diagram. 5

7. a) Define the terms: i) Habitat, ii) Niche and iii) Biome 3
   b) Discuss about different interspecies interactions in the environment. 5
   a) How do interactions among species regulate population dynamics in the ecosystem? 4

8. Write short notes on the following (Any three) 3×4= 12
   a) Ecological Succession
   b) Hydrologic Cycle
   c) Differentiate between habitat and niche
   d) Soil micro-organisms and their roles in the environment.